

Web Application Development

❖ CSS

AMMAR AHMAD KHAN

What is CSS?

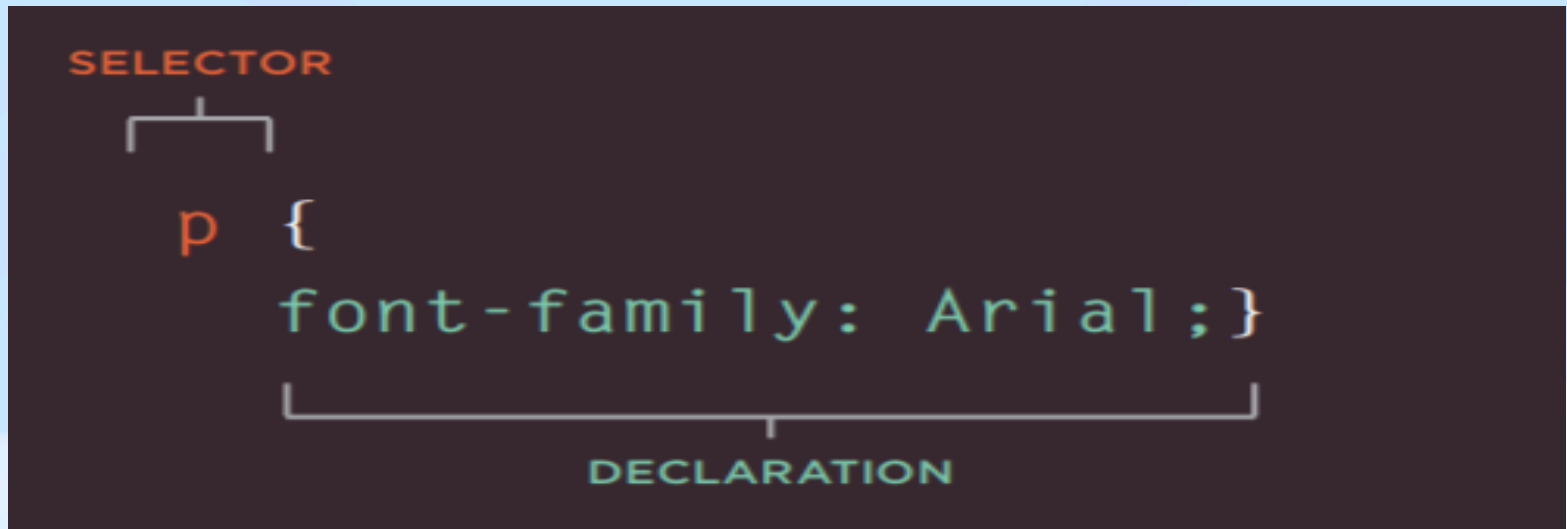
- **Remember:** We use **HTML** to only describe structure and content of a page
 - `<h1>This is heading contents</h1>`
- CSS (Cascading Style Sheets) allows us to apply formatting and styling to the HTML Elements/Tags.
- CSS can control many elements of our web elements:
 - colors, fonts, alignment, borders, backgrounds, spacing, margins, and much more.
- CSS is a style sheet language used to describe the presentation of a document.

CSS

- What is CSS
 - CSS is a document that defines rules for layout of web application frontend.
 - It is designed primarily to enable the **separation** of document content (written in HTML or a similar markup language) from document presentation (written in CSS).
 - CSS can also **allow the same markup page to be presented in different styles** for different rendering methods, such as on-screen, in print, by voice
 - For example,
 - you can specify that the background of the page is cream, all paragraphs should appear in gray using the Arial typeface, or that all level one headings should be in a blue, italic, Times typeface.

CSS

- CSS Rule



- A CSS rule contains two parts: a **selector** and a **declaration**.

- CSS Rule

- The previous rule indicates that all elements should be shown in the Arial typeface.
- Selectors indicate which **element** (in this case p is an element) the rule applies to. The same rule can apply to more than one element if you separate the element names with commas.
- Declarations indicate how the elements referred to in the selector should be styled

CSS

- In a rule, there can be more than 1 element associated with declaration. Example **h1**, **h2**, **h3** in following screen are associated with declaration.

Declaration

```
h1, h2, h3 {  
  font-family: Arial;  
  color: yellow;  
}
```



The diagram shows a CSS rule. The selector 'h1, h2, h3' is on the first line. The opening curly brace '{' is on the second line. The first declaration 'font-family: Arial;' is on the third line, and the second declaration 'color: yellow;' is on the fourth line. The closing curly brace '}' is on the fifth line. A blue arrow points from the word 'Declaration' to the opening curly brace. An orange box highlights the two declarations. Below the first declaration, a bracket labeled 'PROPERTY' spans the text 'font-family:'. Below the second declaration, a bracket labeled 'VALUE' spans the text 'yellow;'.

- This rule indicates that all , **<h1>**, **<h2>**, and **<h3>** elements should be shown in the Arial typeface, and in a yellow color.

CSS

```
h1, h2, h3 {  
    font-family: Arial;  
    color: yellow;}  
    └──┬──┘ └──┬──┘  
    PROPERTY  VALUE
```

- **Properties** indicate the aspects of the element you want to change. For example, color, font, width, height and border.
- Values specify the settings you want to use for the chosen **properties**. For example, if you want to specify a **color** property then the **value** is the color you want the text in these elements to be.

CSS Example

- Example CSS

```
<!DOCTYPE html>
<html>
<body>
  <h1 style="color:blue;text-align:center;">
    This is a heading
  </h1>
  <p style="color:red;">This is a paragraph.</p>
</body>
</html>
```

- Output

This is a heading

This is a paragraph.

CSS

- How to bind CSS with HTML document>
 - Inline
 - styles are placed right where you need them, next to the text you wish to decorate.
 - Internal – Embedded Stylesheets
 - styles are placed at the top of each web page document, before any of the content is listed.
 - On-page <head> style definitions
 - External
 - **External** style sheets are separate files full of CSS instructions. You need to refer this file in your html document.
 - Example
 - <link href="example.css" type="text/css" rel="stylesheet" />

CSS

- Example Inline CSS

```
<!DOCTYPE html>
<html>
<head>
</head>
<body>
  <p style="text-align: center;color: blue;">
    It appears Blue.</p>
  <p id="para1">Me too!</p>
  <p>And me!</p>
</body>
</html>
```

- output

It appears Blue.

Me too!

And me!

CSS

- Example Internal CSS
 - styles are placed at the top of each web page document, before any of the content is listed.

```
<!DOCTYPE html>
<html>
<head>
<style>
  p {
    text-align: center;
    color: blue;
  }
</style>
</head>
<body>
  <p style="text-align: center;color: blue;">It appears Blue.</p>
  <p id="para1">Me too!</p>
  <p>And me!</p>
</body>
</html>
```

It appears Blue.

Me too!

And me!

On-Page Style Definitions(Internal)

- The benefit of this over the previous approach (in line) is that the style definitions are only written once.
- Use this method when you want to apply similar styles to multiple elements on a page, but not on any other pages.
- Embedded style sheets are not scale able to multiple pages.
- To apply the same styles to things on multiple pages, you need to use the next method (Separate style sheets).

CSS

- Example external CSS
 - **External** style sheets are separate files full of CSS instructions. You need to refer this file in your html document.
 - Create a file called ExtCSS.css and paste following

```
body {  
font-family: Arial, Verdana, sans-serif;}  
h1, h2 {  
color: blue;}  
p {  
color: green;}
```

- Create html document and refer ExtCSS.css into it
 - See next slide

CSS

- HTML document with External CSS

```
<!DOCTYPE html>
<html>
  <head>
    <title>Introducing CSS</title>
    <link href="ExtCSS.css" type="text/css" rel="stylesheet" />
  </head>
  <body>
    <h1>This is heading 1</h1>
    <p>a paragraph <i>with italic font</i> from this point onward
    it is not italic</p>
    <h2>How h2 looks like</h2>
    <p>Plants are chosen as much for their functionality
    as for their color and form ... </p>
  </body>
</html>
```

Self closing tag

This is heading 1

a paragraph *with italic font* from this point onward it is not italic

How h2 looks like

Plants are chosen as much for their functionality as for their color and form ...

CSS

- Referring external CSS file
- `<link>`
 - element can be used in an HTML document to tell the browser where to find the CSS file used to style the page. It is an empty element (meaning it does not need a closing tag), and it lives inside the `<head>` element. It should use three attributes:
 - href
 - This specifies the path to the CSS file (which is often placed in a folder called `css` or `styles`).
 - type
 - This attribute specifies the type of document being linked to. The value should be `text/css`.
 - rel
 - This specifies the relationship between the HTML page and the file it is linked to. The value should be `stylesheet` when linking to a CSS file.

Separate Style Sheets

- Advantages of separate style sheets
 - **Reduced file size** - each CSS definition is written only once
 - **Reduced bandwidth** - web browsers will remember (cache) the contents of CSS file
 - **Easier updating** - you only have to change styles in one place for them to change site-wide
 - **Separates the work of styling and creating content** - you can create all your HTML pages first, applying sensible semantic markup, class names and IDs, and then do all your design work later on your style sheets.

CSS Selectors

- In CSS one can select elements for styling purposes based on
 - Type
 - ID
 - Class
 - Attribute

CSS Selectors

- CSS selectors can be divided into different categories
 - Simple selectors
 - Grouping Selector
 - Universal Selector
 - Pseudo-class selectors
 - *Pseudo-elements selectors*
 - *Attribute selectors*

CSS Selectors

- Simple selectors (select elements based on name, id, class)
 - In following, element is selected based on name

```
<!DOCTYPE html>
<html>
<head>
<style>
  p {
    text-align: center;
    color: blue;
  }
</style>
</head>
<body>
  <p>It appears Blue.</p>
  <p id="para1">Me too!</p>
  <p>And me!</p>
</body>
</html>
```

It appears Blue.

Me too!

And me!

CSS Selectors

- Simple selectors (select elements based on name, id, class)
 - In following, element is selected based on Id

```
<html>
<head>
<style>
#para1 {
  text-align: center;
  color: Blue;
}
</style>
</head>
<body>

<p id="para1">Hello World!</p>
<p>This paragraph is not affected by the style.</p>

</body>
</html>
```

Hello World!

This paragraph is not affected by the style.

CSS Selectors

- Simple selectors (select elements based on name, id, class)
 - In following, element is selected based on class

```
<html>
<head>
<style>
.center {
  text-align: center;
  color: Green;
}
</style>
</head>
<body>
```

Green and center-aligned heading

green and center-aligned paragraph.

```
<h1 class="center">Green and center-aligned heading</h1>
<p class="center">green and center-aligned paragraph.</p>

</body>
</html>
```

CSS Selectors

- Simple selectors (select elements based on name, id, class)
 - In following, element is selected based on specific class

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: green;
  font-style: italic;
}
</style>
</head>
<body>
  <h1 class="center">This heading will not be affected</h1>
  <p class="center">This paragraph will be green and center-aligned.</p>
</body>
</html>
```

This heading will not be affected

This paragraph will be red and center-aligned.

CSS Selectors

- Simple selectors (select elements based on name, id, class)

This heading will not be affected

how does it looks lik?

is it green, center-aligned,
large font-size???

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: blue;
}

p.large {
  font-size: 300%;
}
</style>
</head>
<body>

<h1 class="center">This heading will not be affected</h1>
<p class="center">how does it looks lik?</p>
<p class="center large">is it green,
                        center-aligned,
                        and in a large font-size??</p>

</body>
</html>
```

CSS Selectors

- Difference between ID and CLASS

Can not have multiple elements with same *id*
but can have multiple elements with same *class*.

ID *MUST BE UNIQUE*

CSS Selectors

- Universal selector
 - The universal selector (*) selects all HTML elements on the page.

```
<!DOCTYPE html>
<html>
<head>
<style>
* {
  text-align: center;
  color: blue;
}
</style>
</head>
<body>

<h1>rules applied on H1</h1>

<p>On P as well.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
</html>
```

rules applied on H1

On P as well.

Me too!

And me!

CSS Selectors

- Grouping Selector
 - The grouping selector selects all the HTML elements with the same style definitions.

```
<!DOCTYPE html>
<html>
<head>
<style>
  h1, h2, p {
    text-align: center;
    color: brown;
  }
</style>
</head>
<body>
<h1>Hello World!</h1>
<h2>Smaller heading!</h2>
<p>This is a paragraph.</p>
</body>
</html>
```

Hello World!

Smaller heading!

This is a paragraph.

CSS Selectors

- Pseudo-class selectors
 - select elements based on a certain state
 - A pseudo-class is used to define a special state of an element.
 - For example, it can be used to:
 - Style an element when a mouse hover over it
 - Style visited and unvisited links differently
 - Style an element when it gets focus
 - Syntax

```
selector : pseudo-class {  
  property:value;  
}
```

CSS Selectors

- Pseudo-class selectors
 - Anchor Pseudo-classes
 - a:link
 - Describes properties associated with hyperlink
 - a:visited
 - Color of hyperlink once it is visited
 - a:hover
 - Color of hyperlink once user hovers over it
 - a:active
 - Color of hyperlink once clicked

CSS Selectors

- Example

```
<!DOCTYPE html>
<html>
<head>
<style>
  /* unvisited link */
  a:link {
    color: red;
  }

  /* visited link */
  a:visited {
    color: green;
  }

  /* mouse over link */
  a:hover {
    color: hotpink;
  }

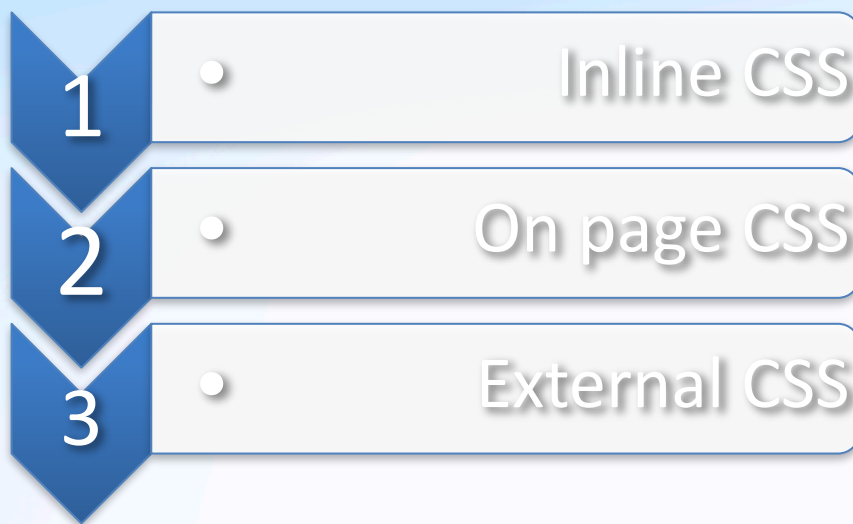
  /* selected link */
  a:active {
    color: blue;
  }
</style>
</head>
<body>
  <p><b><a href="home.html" target="_blank">This is a link</a></b></p>
</body>
</html>
```

Rules for Applying CSS

- If there are two or more rules that apply to the same element, it is important to understand which will take precedence.

CSS Priority Rule

- CSS Properties applies in following order
 - **Inline** properties has highest priority.
 - On page style has less priority then **inline**, and more priority then external CSS.
 - External CSS has least priority.



CSS Properties: Borders

border with solid lines

border with doted lines

```
<html>
<head>
<style>
p.one {
  border-style: solid;
  border-width: 5px;
}

p.two {
  border-style: dotted;
  border-width: thick;
}
</style>
</head>
<body>
  <p class="one"> border with solid lines</p>
  <p class="two"> border with doted lines</p>
</body>
</html>
```

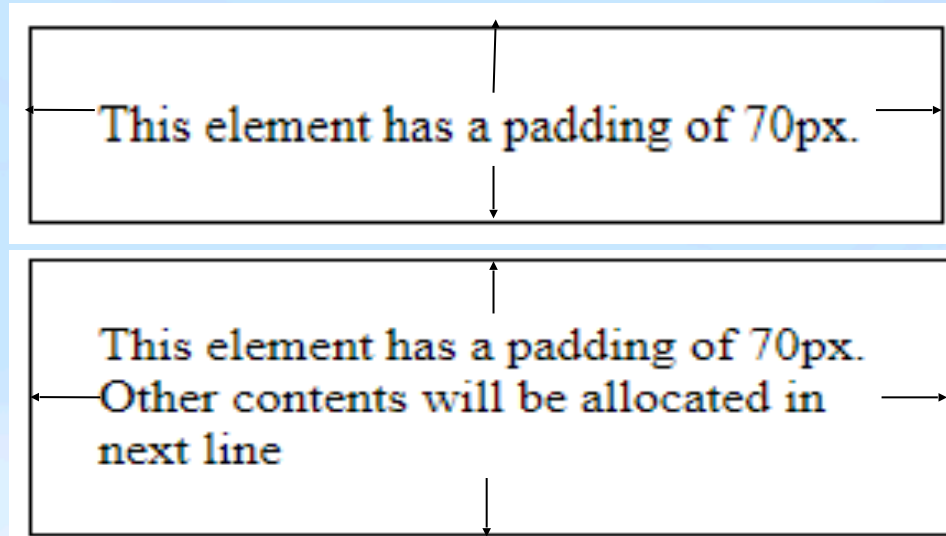

CSS Properties: Borders

border with solid lines

```
<html>
<head>
<style>
p.one {
  border-style: solid;
  border-width: 5px 10px 15px 25px; //top right bottom left
}
</style>
</head>
<body>
  <p class="one"> border with solid lines</p>
</body>
</html>
```

CSS Properties: Page Padding

- Example



CSS Properties: Page Padding

- Page Padding

```
<html>
<head>
<style>
div {
  padding: 20px;
  border: 1px solid;
}
</style>
</head>
<body>
```

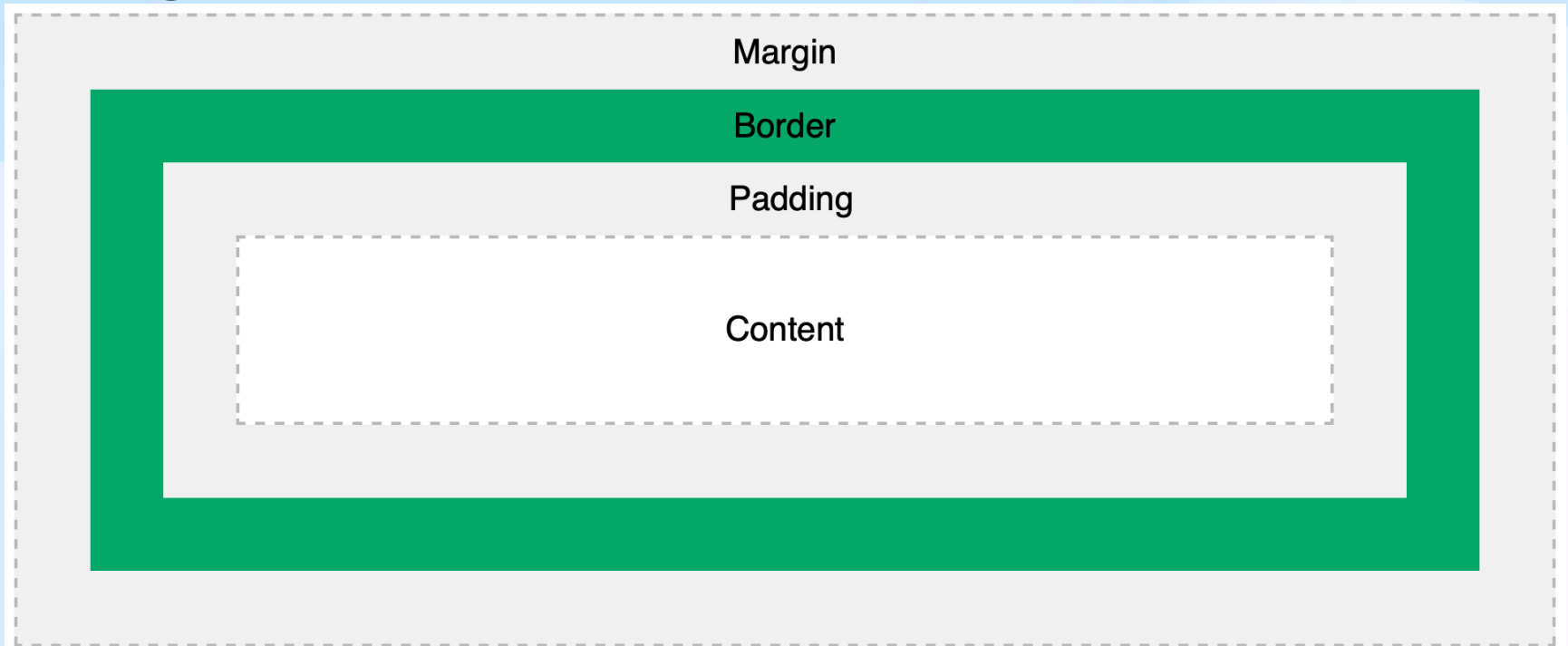
```
<div>This element has a padding of 70px. Other contents will
be allocated in next line</div>
```

```
</body>
</html>
```

This element has a padding of 70px.
Other contents will be allocated in
next line

Box Model

- The CSS box model is essentially a box that wraps around every HTML element. It consists of: content, padding, borders and margins. The image below illustrates the box model:



Explanation of Diff Parts:

Content - The content of the box, where text and images appear

Padding - Clears an area around the content. The padding is transparent

Border - A border that goes around the padding and content

Margin - Clears an area outside the border. The margin is transparent

Explanation of Diff Parts:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  background-color: lightgrey;
  width: 300px;
  border: 15px solid green;
  padding: 50px;
  margin: 20px;
}
</style>
</head>
<body>
```

```
<h2>Demonstrating the Box Model</h2>
```

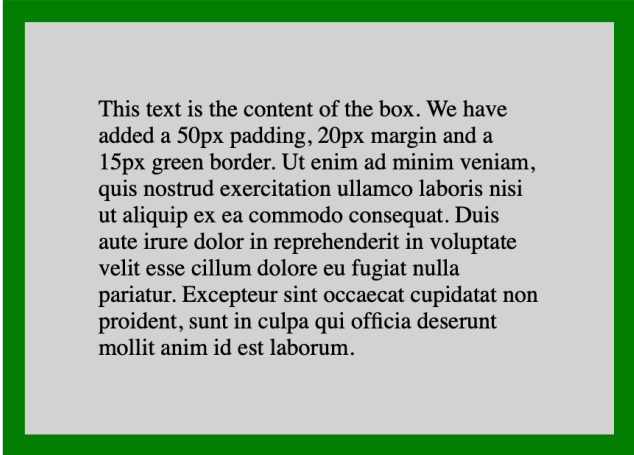
```
<p>The CSS box model is essentially a box that wraps around
every HTML element. It consists of: borders, padding, margins,
and the actual content.</p>
```

```
<div>This text is the content of the box. We have added a 50px
padding, 20px margin and a 15px green border. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut
aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in
culpa qui officia deserunt mollit anim id est laborum.</div>
```

```
</body>
</html>
```

Demonstrating the Box Model

The CSS box model is essentially a box that wraps around every HTML element. It consists of: borders, padding, margins, and the actual content.



This text is the content of the box. We have added a 50px padding, 20px margin and a 15px green border. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

IMAGES

```
<head>
<style>
img.align-left {
    float: left;
    margin-right: 10px;}

img.align-right {
    float: right;
    margin-left: 10px;}

img.medium {
    width: 100;
    height: 100px;}

img.small {
    width: 75;
    height: 75px;}
</style>
</head>
```

```
<body>

<p>

<b><i>Magnolia</i></b> is a large genus that
contains over 200 flowering plant species...</p>

<p>

Some magnolias, such as <i>Magnolia stellata</i>
and <i>Magnolia soulangeana</i>, flower quite
early in the spring before the leaves open...</p>

</body>
```



Magnolia is a large genus that contains over 200 flowering plant species...

Some magnolias, such as *Magnolia stellata* and *Magnolia soulangeana*, flower quite early in the spring before the leaves open...



IMAGES

```
<head>
<style>
img.align-left {
  float: left;
  margin-right: 10px;}

img {
  opacity: 0.5;
  filter: alpha(opacity=50);
}
</style>
</head>
```

Image with 50% opacity:



Magnolia is a large genus that contains over 200 flowering plant species...

Some magnolias, such as *Magnolia stellata* and *Magnolia soulangeana*, flower quite early in the spring before the leaves open...

```
<body>

<p>Image with 50% opacity:</p>
  

<b><i>Magnolia</i></b> is a large genus that
contains over 200 flowering plant species...</p>

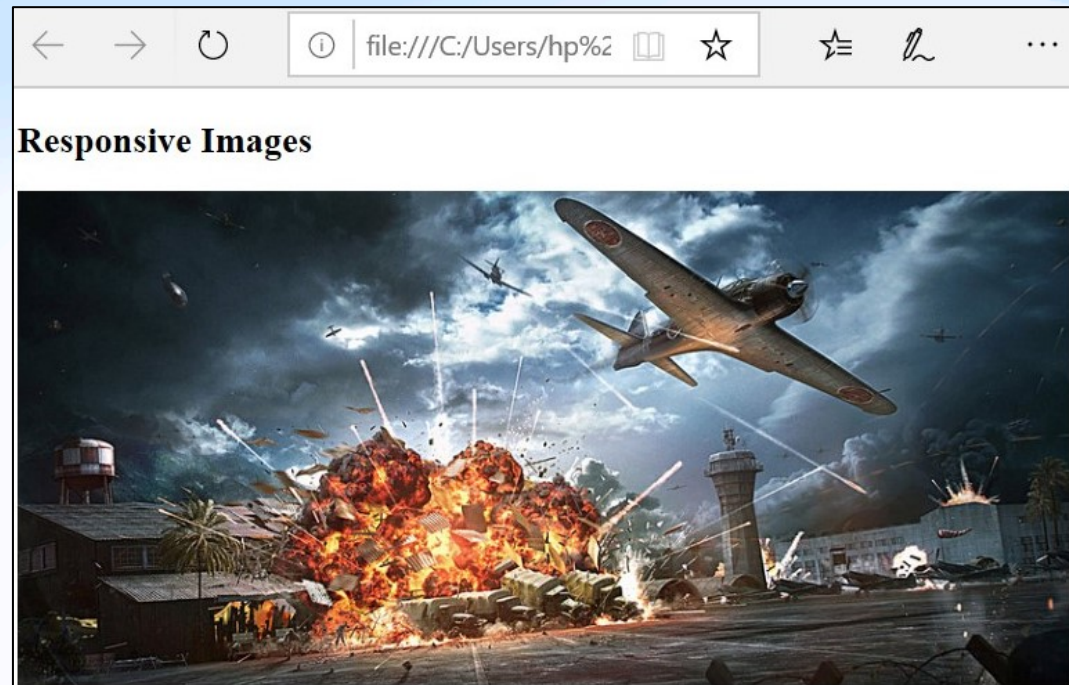
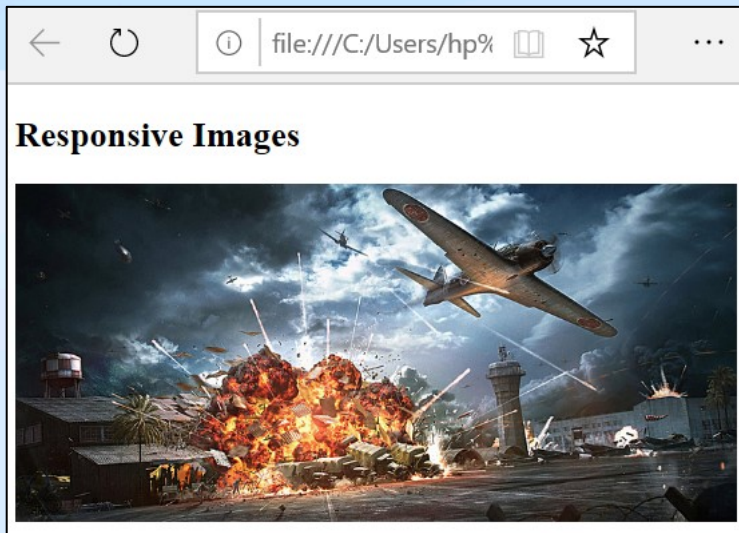
Some magnolias, such as <i>Magnolia stellata</i>
and <i>Magnolia soulangeana</i>, flower quite
early in the spring before the leaves open...</p>

</body>
```


IMAGES

```
<head>
<style>
img {
  max-width: 100%;
  height: auto;
}
</style>
</head>

<body>
<h2>Responsive Images</h2>
  
</body>
```



Task

```
<!DOCTYPE>
<html>
  <head>
    <title>Table Properties</title>
    <style type="text/css">
      body {
        font-family: Arial, Verdana, sans-serif;
        color: #111111;}
      table {
        width: 600px;}
      th, td {
        padding: 7px 10px 10px 10px;}
      th {
        text-transform: uppercase;
        font-size: 90%;
        border-bottom: 2px solid #111111;
        border-top: 1px solid #999;
        text-align: left;}
      tr.even {
        background-color: #efefef;}
      tr:hover {
        background-color: #c3e6e5;}
      .money {
        text-align: right;}
    </style>
  </head>
```

Task

```
<body>
  <h1>First Edition Auctions</h1>
  <table>
    <tr>
      <th>Author</th>
      <th>Title</th>
      <th class="money">Reserve Price</th>
      <th class="money">Current Bid</th>
    </tr>
    <tr>
      <td>E.E. Cummings</td>
      <td>Tulips & Chimneys</td>
      <td class="money">$2,000.00</td>
      <td class="money">$2,642.50</td>
    </tr>
    <tr class="even">
      <td>Charles d'Orleans</td>
      <td>Poemes</td>
      <td class="money"></td>
      <td class="money">$5,866.00</td>
    </tr>
    <tr>
      <td>T.S. Eliot</td>
      <td>Poems 1909 - 1925</td>
      <td class="money">$1,250.00</td>
      <td class="money">$8,499.35</td>
    </tr>
    <tr class="even">
      <td>Sylvia Plath</td>
      <td>The Colossus</td>
      <td class="money"></td>
      <td class="money">$1031.72</td>
    </tr>
  </table>
</body>
</html>
```

Task

- Output

First Edition Auctions

AUTHOR	TITLE	RESERVE PRICE	CURRENT BID
E.E. Cummings	Tulips & Chimneys	\$2,000.00	\$2,642.50
Charles d'Orleans	Poemes		\$5,866.00
T.S. Eliot	Poems 1909 - 1925	\$1,250.00	\$8,499.35
Sylvia Plath	The Colossus		\$1031.72